main.c

Run ▶

```c
#include <stdio.h>

int main()
{
    char op, arg1, arg2, result;

    printf("Enter an expression (e.g., a+b): ");
    scanf(" %c %c %c %c", &result, &op, &arg1, &arg2);

    printf("\nTarget Code:\n");

    printf("MOV R0, %c\n", arg1);

    if (op == '+')
        printf("ADD R0, %c\n", arg2);
    else if (op == '-')
        printf("SUB R0, %c\n", arg2);
    else if (op == '*')
        printf("MUL R0, %c\n", arg2);
    else if (op == '/')
        printf("DIV R0, %c\n", arg2);
    else
    {
        printf("Invalid operator.\n");
        return 1;
    }

```

Output

```
Enter an expression (e.g., a+b): x + a b

Target Code:
MOV R0, a
ADD R0, b
MOV x, R0
```

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```
6
7       printf("Enter an expression (e.g., a+b): ");
8       scanf(" %c %c %c %c", &result, &op, &arg1, &arg2);
9
10      printf("\nTarget Code:\n");
11
12      printf("MOV R0, %c\n", arg1);
13
14      if (op == '+')
15          printf("ADD R0, %c\n", arg2);
16      else if (op == '-')
17          printf("SUB R0, %c\n", arg2);
18      else if (op == '*')
19          printf("MUL R0, %c\n", arg2);
20      else if (op == '/')
21          printf("DIV R0, %c\n", arg2);
22      else
23      {
24          printf("Invalid operator.\n");
25          return 1;
26      }
27
28      printf("MOV %c, R0\n", result);
29
30      return 0;
31  }
32
```

Output

```
Enter an expression (e.g., a+b): x + a b

Target Code:
MOV R0, a
ADD R0, b
MOV x, R0
```